MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

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***Abstract*—We propose MinImageScorer, an error-driven framework that scores per-object detection difficulty from baseline model , combining confidence and localization penalties calibrated via Pearson correlation optimization. Validated on MinneApple and Global Wheat Head (GWHD), the score reliably ranks objects and images by difficulty (Pearson correlation up to 0.82 with miss rate) and correctly identifies domain- and size-level failure patterns without supervision. This score was then used to guide copy-paste augmentation, which outperforms random selection on MinneApple (ΔAP=$+0.013$) but both strategies fall below baseline due to distributional shift. On GWHD, domain shift renders copy-paste ineffective under either strategy. These results show that MinImageScorer correctly identifies hard objects, and that copy-paste can only work with a very limited cased.**

(*Does this abstract sound too straight and hard to understand ? If it is, make it better. do not change the length too much)

***Index Terms*—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift**

I. Introduction

Agricultural object detection benchmarks such as MinneApple [1] and Global Wheat Head (GWHD) [2] exhibit distinct structural failure modes: MinneApple is dominated by sub-resolution small objects, while GWHD contains severe cross-domain visual variation across geographic sources. Copy-paste augmentation has demonstrated gains on MS COCO [3], [4], where objects are well-distributed and within-distribution. However, the method has not been thoroughly evaluated on structurally constrained datasets like agricultural benchmarks, where applying copy-paste blindly risks corrupting the training distribution rather than improving it. No prior work provides an empirical framework to diagnose when copy-paste is beneficial before augmentation is applied. (Is this gap addressing good ? If not do it better, i think we can make it more clear and direct, and also more concise agriculture dataset have unique failure modes (small objects, domain shift,..) I still do not sure about how to connect the gap with the proposed contribution the contribution is the score that can identify the failure mode. It was hard to say that using the score to guide copy-paste improve a result because althought the ap score is increased compared to random but it is still below the baseline. and even for gwhd the score-guided was not even better. We can explain this usign the experiment about the feature space when just copy and paste has already shift the data too far away from the test distribution. Improve this part for me.)

We address this gap with three contributions: (1) **MinImageScorer** (MIS), a per-object difficulty score derived from model failure signals with data-driven weight calibration; (2) empirical validation showing the score correctly identifies image-, domain-, and object-size-level difficulty without access to test labels; (*you can think and merge contribution 1,2 together)and (3) a controlled study demonstrating that score-guided copy-paste outperforms random selection on resolution-limited datasets but both strategies fail on domain-shifted datasets.

II. Methodology

A. Scoring Formulation

Let $G(I)$ be ground-truth objects in image I and $\mathcal{P}(g) = \{p : \mathrm{IoU}(p, g) \geq \tau\}$ the matched predictions. Denoting $c_p = \mathrm{conf}(p)$ and $u_p = \mathrm{IoU}(p, g)$, the *object difficulty score* is:

$$S_{\mathrm{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} \left[\alpha(1 - c_p) + \beta(1 - u_p)\right], & \mathcal{P}(g) \neq \varnothing, \\ \alpha + \beta, & \text{otherwise}, \end{cases} \tag{1}$$

where α and β weight confidence and localization penalties. Missed objects receive the maximum score $\alpha + \beta$. The *image-level score* combines mean object difficulty with false-positive rate:

$$S_{\mathrm{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\mathrm{obj}}(g) \; + \; w_2 \cdot \mathrm{FPrate}(I). \tag{2}$$

Object scoring uses low-confidence predictions (τ=0.01) to capture near-misses; FP rate uses optimal-threshold predictions to reflect deployment errors. Weights α, β, w_1, w_2 are calibrated by maximizing Pearson correlation with per-image miss rate and FP rate on the training set, requiring no domain-specific assumptions.

B. Score-Guided Copy-Paste Pipeline

A baseline detector (YOLOv8s) is trained with all built-in augmentations disabled. Inference on the training set yields confidence and IoU signals used to compute S_{obj} and S_{img}. For score-guided augmentation, objects are sampled proportionally to S_{obj} via exponential weighting; for random, sampling is uniform. Both conditions paste 2–8 objects per augmented

TABLE I

PEARSON CORRELATION OF S_{img} WITH PER-IMAGE MISS RATE AND FP RATE (TRAINING SET, WITH TRADITIONAL AUGMENTATION).

Dataset	r(score, miss)	r(score, FP)	Inter-seed r
MinneApple	0.698	0.688	0.806
GWHD	0.816	0.818	0.900

image within the source image and retrain under identical hyperparameters.

(*Explain more about this part, I think it too short and not clear enough for reader to image what i do with the score guided copy and paste.)

III. Score Validation

A. Correlation with Detection Failures

Table I reports Pearson correlation between S_{img} and per-image miss rate and FP rate on training images. All cases show strong positive correlation ($r \geq 0.69$), indicating the score reliably captures per-image difficulty across all conditions. The strongest relationship (GWHD with traditional augmentation, r=0.82) suggests the score accurately identifies hard images when the baseline model is stable. The inter-seed correlation remains high across all cases (mean $r \geq 0.80$), confirming that domain-specific factors drive the score more than training noise.

(*For this experiment i think just keep the the correlation with missrate and false positive rate is enough)

B. Domain- and Size-Level Validity

Beyond per-image correlation, the score recovers structural difficulty without access to test labels. On GWHD, mean S_{img} exhibits perfect inverse rank correlation with domain-level AP across all seven source domains (Spearman ρ=-1.00, p<0.001). All seven domains rank identically when ordered by AP and by score: the hardest domain (*arvalis_2*, AP = 0.365) scores 0.424, while the easiest (*rres_1*, AP = 0.572) scores only 0.200. This alignment is recovered from training-set model failures alone, without test labels (Table II). On MinneApple, small objects (≤32 px) average a score of 0.378 vs. 0.222 for medium objects, exactly matching the AP gap (0.132 vs. 0.522). Together, these results confirm that MIS captures image-, domain-, and object-size-level difficulty from training-set signals without supervision.

(for this part i want to show case the explain the table 2 and one more picture that show the correlation of the score with the size of object on minneapple and the score with the domain on gwhd. State that: the auther of minneapple and gwhd show that the most difficult part of this dataset is the small object and the domain shift respectively, based on the result we just show case, we can say that the score can capture this kind of difficulty without any more information.)

TABLE II

GWHD DOMAIN DIFFICULTY: MEAN SCORE VS. BASELINE AP. ALL 7 DOMAINS RANK IDENTICALLY (SPEARMAN ρ=-1.00, p<0.001, N=7 DOMAINS).

Domain	**n img**	**AP**	**AP$_{50}$**	**Mean S$_{\text{img}}$**
rres_1	363	0.572	0.962	0.200
ethz_1	592	0.550	0.940	0.227
inrae_1	146	0.523	0.954	0.251
arvalis_3	457	0.466	0.918	0.294
usask_1	153	0.408	0.874	0.312
arvalis_1	827	0.403	0.890	0.331
arvalis_2	162	0.365	0.813	0.424

IV. Augmentation Experiments

A. Setup

All models use YOLOv8s initialized from COCO pretrained weights. Training: MinneApple 200 epochs / patience 10; GWHD 100 epochs / patience 8; batch 16; input 1024×1024; lr 10^{-2}. All built-in augmentations are disabled to isolate copy-paste effects. Augmentation ratio 0.3; scale jitter $[0.9, 1.1]$ (MA) and $[0.8, 1.2]$ (GWHD). Do not need to specifiy the hyperparamater too much. focus more on teh augmentation strategy. Of we scale the image and rotate it rampdomly We do that because the researh - cite it show that this is a good way of doing copy and paste augmentation.

B. Results

Table III shows that on MinneApple, score-guided copy-paste outperforms random across all sub-metrics (ΔAP=+0.013), confirming the score directs augmentation toward higher-signal objects. However, both conditions fall below baseline (-0.024 for Score CP). The AP$_{75}$ drop (0.452→0.440) indicates degraded localization precision from pasting hard objects without blending; the AP$_S$ drop (0.300→0.286) indicates score-selected objects are predominantly sub-resolution and cannot be made learnable by repetition alone.

On GWHD, score-guided and random selection are indistinguishable (ΔAP=-0.002) and both fall below baseline (-0.023). Score-guided sampling concentrates objects from domain-minority sources (*arvalis*), amplifying existing domain imbalance. Backbone feature analysis (YOLOv8s P4, UMAP) confirms that augmented images are displaced from the test distribution on both datasets, with greater displacement on GWHD, directly explaining the consistent degradation.

V. Conclusion

MinImageScorer reliably quantifies detection difficulty from model failure signals, validated by strong per-image correlations (r up to 0.82 with miss rate) and perfect domain-level rank recovery (Spearman ρ=-1.00 across all 7 GWHD domains without test labels). Object-size difficulty is correctly ordered on MinneApple. Score-guided copy-paste outperforms random selection on MinneApple (+0.013 AP) but both strategies

TABLE III
THREE-SEED MEAN COCO AP. **BOLD** = BEST COPY-PASTE CONDITION PER DATASET.

Dataset	Condition	AP	AP_{50}	AP_{75}	AP_S	AP_M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	**0.428**	**0.759**	**0.440**	**0.286**	**0.581**
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	**0.484**	**0.905**	**0.455**	**0.098**	**0.477**

degrade below baseline due to distributional shift between augmented and test images. On GWHD, domain shift makes copy-paste ineffective regardless of selection quality.

The practical implication is direct: copy-paste is effective when objects are well-represented in feature space (COCO-like underrepresentation), but fails when the failure mode is structural (sub-resolution limits on MinneApple or domain shift on GWHD). MIS provides the diagnostic criterion: inspect the size and domain distribution of high-score objects before applying copy-paste augmentation.

Future work should explore two directions: (1) *Domain-aware pasting*: select pasted objects only from training domains similar to the test distribution, reducing the distributional mismatch that causes degradation here; and (2) *Dynamic augmentation*: compute object selection during training rather than fixing scores upfront, allowing the method to adapt as the model improves. Both approaches aim to keep augmented images within the test distribution, addressing the core failure mode identified in this work.

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